

Derbyshire Shared Care Pathology Guidelines

Hypophosphataemia in Adults

Importance of Hypophosphataemia

Severe hypophosphataemia causes muscle weakness, which may lead to respiratory impairment

When is hypophosphataemia considered a medical emergency?

- Phosphate 0.80 – 0.48 mmol/L Mild – Not usually a medical emergency
- Phosphate 0.48 – 0.32 mmol/L Possible medical emergency
- Phosphate < 0.32 mmol/L Severe - Likely medical emergency



When the phosphate is <0.32 mmol/L the result will usually be telephoned to Derbyshire health unit /111 when the GP practice is closed.

When is hypophosphataemia suspected?

Mild hypophosphataemia: is much more common and is usually not associated with clinical signs and symptoms, but rickets or vitamin D resistant osteomalacia can develop in chronic hypophosphataemia.

Severe hypophosphataemia: often arises in patients who are critically ill from other causes, and the symptoms of phosphate deficiency (such as severe muscle weakness, acute respiratory failure, decreased cardiac output, tissue hypoxia, haemolysis, confusion, irritability and coma) may mimic features of these illnesses.

What are the likely causes in this patient?

- Primary hyperparathyroidism
 - PTH increases phosphate excretion by the kidneys
- Vitamin D deficiency
 - Reduced phosphate absorption from the gut and secondary hyperparathyroidism
- Diarrhoea and vomiting
 - Prolonged decreased intestinal phosphate absorption
- Marathon running or prolonged intense exercise
 - Increased muscle uptake of phosphate to replenish creatine phosphate stores

CHISCP64: Hypophosphataemia in adults, Revision No 1

Expiry Date: 31st Jan 2027

Authors: Dr A Ugur Consultant Endocrinologist UHDB, Dr A Pillai Consultant Endocrinologist CRH, Dr R Robinson Consultant Endocrinologist CRH, Dr P Blackwell GP at Hannage Brook medical practice, Ms S Knowles Consultant Clinical Scientist UHDB.

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- Post i.v. iron infusion (Ferinject)
 - Temporary change in hormone balance
- Alcohol withdrawal
- Ingestion of non-absorbable antacids (e.g. aluminium hydroxide)
 - Phosphate absorption is prevented in the gut
 - Now rare due to availability of alternatives for peptic ulcer disease
- Re-feeding of malnourished patients
 - Movement of phosphate into cells
- Liver failure

Rare Causes of hypophosphatemia

- Recovery from hospital care with
 - Severe burns, Salicylate poisoning, Alcoholic ketoacidosis, Sepsis and bacteraemia
- Congenital defects of tubular phosphate reabsorption
 - Fanconi syndrome
 - Autosomal dominant hypophosphataemic rickets
 - X-linked hypophosphataemic rickets
 - Oncogenic osteomalacia
 - Hereditary hypophosphataemic rickets with hypercalciuria

What action should be taken?

Repeat the phosphate to confirm the low result on a fasting sample (insulin reduces serum phosphate).

Mild hypophosphataemia

If the cause is known, treatment of the underlying disorder, possibly with oral phosphate supplementation.

If the cause is not known and hypophosphataemia is persistent, these initial investigations below are appropriate:

- Serum PTH, Vitamin D, U&E, phosphate, calcium

Address possible causes as identified by the above investigations (refer to the following guidelines): [Primary Hyperparathyroidism](#), [vitamin D replacement](#)

- There is no need to replace phosphate where PTH dependent hypercalcaemia is the cause.

If all the investigations above are normal, the cause has not yet been identified, and the mild hypophosphataemia is persistent on two tests 2 months apart, please consider the following:

- Is the hypophosphataemia acquired or long-standing (based on historic readings)?
- Is there a family history of hypophosphatemia or rickets?
- Does the patient have known fibroma or new skin or subcutaneous lesion?

Only when the cause of mild hypophosphatemia is unknown and due to high urinary phosphate excretion would further advice and possible onward investigations be necessary.

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High urinary phosphate excretion is identified by sending a paired urine and serum for phosphate and creatinine for the laboratory to calculate TmP/GFR and guidance on whether it is inappropriately low.

- TmP/GFR is the theoretical lower limit of serum phosphate below which all filtered phosphate would be reabsorbed.
- A lower TmP/GFR is inappropriate in hypophosphataemia when all filtered phosphate should be reabsorbed.

1. Overnight fast
2. The first voided urine should be discarded.
3. Collect a 25 ml sample of the second void urine into a universal container for urine creatinine and phosphate
4. Send with serum electrolytes and phosphate
5. Laboratory will perform calculation of significance with appropriate comment regarding the interpretation of the result as compared to age and gender specific references.

Please either perform these fasted paired samples without phosphate replacement or with information included about the phosphate supplements if essential. The result will not be returned if the serum phosphate is normal.

If there is history consistent with acquired hypophosphataemia with low TmP/GFR or congenital hypophosphatemia, please refer to Endocrinology for further assessment. If TmP/GFR is not suggestive of renal phosphate losses, manage phosphate symptomatically encouraging a diet higher in phosphate.

Symptomatic hypophosphataemia

If your patient has any of the severe symptoms listed above, regardless of the phosphate level it is likely your patient will need acute medical assessment, therefore consider referring to acute medical team as appropriate.

Contacts

UHDB-RDH

Duty Biochemist	01332 789383 (8am to 7pm, Mon – Fri)
On Call Consultant Biochemist	Via RDH switchboard, 01332 340131 (24/7)
Endocrinology Advice	07879 115507 (9am – 5pm, Mon – Fri)
Renal Registrar (9am to 9pm)	via RDH switchboard bleep 8121
GP Phone Triage	01332 789099
MAU and ACC	01332 788707 <u>OR</u>
MAU Nurse in Charge	07917 650751

UHDB-QHB

Duty Biochemist	01283 593313 (8am to 7pm, Mon – Fri)
Endocrinology advice	Via endocrine secretaries 01283 593259

Chesterfield

Duty Biochemist	01246 512212 (9am to 5pm, Mon – Fri)
On Call Consultant Biochemist	Via CRH switchboard, 01246 277271
A&E, EMU and CDU	Via CRH switchboard

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